Early Prediction of Student Outcomes in Online Learning Using OULAD: A Leakage-Aware Comparison of Traditional and Deep-Learning Models

Project Report

Author: Iriza Kate Lorna Julius

Abstract

In this paper, we propose a multiclass classification approach to predict the Pass, Fail and Withdrawn outcomes in the Open University Learning Analytics Dataset (OULAD) using an early-warning framework. Only information available by day 56 of a module presentation is used, allowing for task to be inserted before the end of course. All records for a student are placed in the same train, validation or test partition to decrease leakage. The study investigates the performance of logistic regression, random forests, histogram gradient boosting, a regularised multilayer perceptron and a hybrid LSTM-tabular network. Validation macro-F1 is used to rank performance and is evaluated on a held-out test set using macro-F1, balanced accuracy, ROC-AUC, precision-recall curves, per-class recall, bootstrap confidence intervals and error analysis. The selected traditional model gives test macro-F1 = 0.5489, and accuracy = 0.6664, the selected MLP gives test macro-F1 = 0.5412, and accuracy = 0.6347. A traditional model is found to have higher overall accuracy, and MLP has higher accuracy on Withdrawn students (recall = 0.4072 versus 0.3366). According to paired bootstrap analysis, there is no statistically significant difference between the macro-F1 and ROC-AUC of the models, suggesting that they are quite similar with their performance differing only slightly on the precision-recall trade-off. The assessment and VLE engagement data is found to be contributing significantly beyond the demographic context through feature ablation. The findings support the application of the models as a human-in-the-loop early-warning system and not an automatic decision maker.

Index Terms: learning analytics, early warning systems, student outcome prediction, OULAD, machine learning, deep learning, class imbalance, VLE clickstream, dropout prediction.

Project Resources

GitHub repository: https://github.com/k10-j/OULAD_Early_Warning_Classification

Google Drive folder: ML_Final_Project_OULAD_Early_Warning

Project video: **https://youtu.be/-yoODRuw4gM**

OULAD dataset: https://research.stem.open.ac.uk/ouanalyse/dataset/

I. Introduction

Learning analytics has shifted from reporting on past learning to predictive systems that enable identification of students who may need help before it is too late. This is especially important in online and blended learning, where a virtual learning environment (VLE) produces detailed records of students' participation, assessment activity and access to learning resources. In order to make an effective early-warning prediction, in addition to the correct classification, it is also necessary to control the information available during the prediction.

This study proposes and tests a multiclass early-warning classifier with the use of the Open University Learning Analytics Dataset (OULAD). OULAD is a fusion of the demographic data, assessment history, data on how students interact with the VLE, and final student achievement in Open University module presentations. The original data set consists of 32,593 student registrations, 22 course presentations and over ten million daily click records on the VLE [1]. In this study Distinction is merged with Pass, which gives three outcome classes: Pass, Fail, Withdrawn.

At the end of the eighth week (day 56) of the presentation of a module, the study concentrates on predicting the final outcome (the 79th week, day 556). This temporal constraint would allow the models to only be based on

information that would reasonably have preceded an early intervention. The records of the information that wasn't available at this prediction point were therefore excluded.

The study is based on the application of conventional machine learning and deep learning, such as a regularised multilayer perceptron (MLP) and a hybrid LSTM-tabular model. Model performance is assessed by macro-F1, balanced accuracy, ROC-AUC, precision-recall performance and confusion matrices, specifically focusing on the minority Fail and Withdrawn classes.

The best traditional model performed 0.5489 for macro-F1 and 0.6664 for accuracy and the best MLP model was 0.5412 for macro-F1 and 0.6347 for accuracy. The MLP performed better on Withdrawn recall (0.4072 vs 0.3366), however. These findings illustrate the potential for accurate prediction of outcomes early in the school year, but also the need for a balance between accuracy and sensitivity to identifying at-risk students.

II. Literature Review

Learning analytics refer to the process of gathering, analysing, and interpreting data that is created during learning to gain insight into students' behaviors and enhance learning outcomes [2]. The growing prevalence of digital learning environments is leading to a greater reliance on data-driven approaches to provide schools with information about students who are at risk for dropping out or failing. A goal of predictive modelling is to help achieve this objective by making predictions based on student historical and behavioural data to guide the provision of student support [3]–[5].

The ability to identify patterns in relation to academic performance has been shown to be viable in previous research using machine learning. Previous academic performance, student demographics, and test scores and engagement with online learning platforms are predictive signals that can be useful [3]–[5]. But, performance in predicting is not sufficient to prove the usefulness of a model on educational outcomes. Recognizing students that are at risk is different from intervening effectively to support those students. Learning analytics can be used as a tool to guide interventions, but the utility of prediction is highly dependent on interpretation and utilization of the prediction [4], [6].

The Open University Learning Analytics Dataset (OULAD) is a helpful tool in studying student performance in online and distance learning. It can include student demographics, course enrollments, test scores, and how often students log in to the Virtual Learning Environment (VLE) and interact with it, including how many times they click on the website each day [1]. These supplementary data resources provide an opportunity to explore the connections between student characteristics, assessment practices, online engagement, and student achievement. In particular, VLE interaction data provide the possibility to examine not only levels of engagement but also the changing nature of engagement over time.

However, there are a number of challenges in the methodology of student outcome prediction given these opportunities. Firstly, predictive models should not include information that would not be available at the time of the prediction. Incorporating data collected beyond the prediction point can result in a temporal form of data leakage, and lead to over-optimistic performance estimates. Secondly, academic results are frequently not even across classes. Modelling that is optimised to maximise accuracy on the majority class may perform well on those classes, and not perform well on classes of students where they are less prevalent but still important, such as students who drop out. To this end, complementary metrics like macro-averaged F1 score, balanced accuracy and class-specific precision and recall [5] [7] should be taken into account when evaluating the results.

Other work has examined the application of sequences of engagement and mixed modelling techniques to represent temporal changes in student engagement. Sequential features can specify trends in engagement measured over consecutive weeks instead of providing a total sum. The research in the literature on clickstream and hybrid studies give some grounds for exploring the impact of integrating sequential engagement with structured student data on prediction [7]. But the comparison of models should only be performed if the data is consistently partitioned, leakage-free feature building, and testing for all outcome classes are performed to determine if added model complexity is providing meaningful improvements.

This work follows these guidelines and evaluates three different models (OULAD based conventional machine learning models, multilayer perceptron and hybrid LSTM–tabular model). Predictions are limited to information available on day 56, and student-level grouped splitting to minimize leakage between training, validation, and test

sets. The validation macro-F1 is considered for model selection, and overall performance and class-specific results for held-out data are considered. The study compares and contrasts sequential modelling with traditional methods under a similar temporal evaluation design to provide a controlled comparison relevant to the identification of students who are at risk for failure or drop out.

III. Methodology

The Open University Learning Analytics Dataset (OULAD) was used to study supervised multiclass classification. The raw tables were added at the student-module-presentation level, and all predictors limited to the information available at day 56. Distinction with Pass was combined, and the original outcomes – Pass, Distinction, Fail and Withdrawn – were consolidated to yield three classes. After temporal eligibility filtering, there were 26,513 records left with about 58% Pass, 27% Fail and 15% Withdrawn. Class imbalance was the reason for class weighting, and for macro-averaged evaluation metrics.

Records for the same student were put into a single partition, so as to avoid data loss. Grouped stratification by student identifier resulted in sets of training and validation records each representing about 69.7% of all records, and test records representing about 15.1% of all records. The test set was not altered in the selection of the model to give an independent estimate of generalisation.

The feature engineering was based on demographic and contextual data, assessment behaviour and virtual learning environment (VLE) engagement. Academic variables were used that included previous attempts, credits taken, assessment scores, credits submitted, and lateness, and demographic variables captured age band, education, and disability status, and region. Engagement metrics consisted of total clicks, active days, number of distinct resources, the types of resources used, and patterns of interaction per week. Other variables were related to engagement dynamics, such as number of weeks engaged, clicks per week of active use, days since last use, pre-course engagement, and engagement variation over time. The weekly click, active days and distinct-resource counts were also preserved as a sequence of eight weeks to be used for neural models.

To preprocess the data, missingness indicators were added, missing values were imputed by median, count-type variables were log-transformed and scaled variables were standardised. One-hot encoding, ordinal encoding and integer encoding with learned embeddings were used for categorical variables in conventional models and neural networks respectively. Training data were used to fit imputation parameters, to apply scaling transformations, and to create cohort-relative engagement references; all of these are then applied in the same way to the validation and test sets.

The modelling experiments involved comparing a majority class baseline, logistic regression, random forests and histogram gradient boosting with deep-learning approaches. The neural architectures consisted of a multilayer perceptron (MLP) comprising dense blocks of 128 and 64 ReLU units, and a hybrid form of LSTM-tabular model that featured categorical embeddings and an eight-week behavioural sequence. The balanced class weights, dropout rate of 0.3 and L2 regularisation with (10^{-4}) were used. The neural models were trained with Adam optimizer, sparse categorical cross-entropy loss, and early stopping to prevent overfitting with a batch size of 256. Traditional models were tested using various class-weighting methods, hyperparameter optimization, feature-group ablation and probability-based decision-rule adjustment.

Only the validation macro-F1 was used for model selection. To get the results of the comparison, EXP-09 was chosen as the best traditional model and EXP-11 was chosen as the best deep-learning model. The held-out test set was used to evaluate their final performance with accuracy, balanced accuracy, macro-F1, ROC-AUC, precision-recall AUC, and precision and recall per class. The classification errors and minority-class detection were analyzed using confusion matrices and one-versus-rest ROC and precision-recall curves.

Confidence intervals and paired comparisons of the selected models were provided using 1,000 student-level bootstrap resamples of the data. The learning curves were studied to evaluate the model convergence and generalisation. Further analyses examined performance by module, by presentation and by demographic subgroups, and permutation importance examined the importance of individual features and larger groups of features. These procedures together gave a systematic assessment of predictive performance and generalisation, as well as practical trade-offs with respect to early-warning classification.

IV. Results

The best validation macro-F1 score was obtained for the traditional models with EXP-05 (weighted histogram gradient-boosting model) with macro-F1 = 0.5480 and regularised random forest with macro-F1 = 0.5471. A longer fitting time of ~116 s was observed, yet we did not see any improvement in the result for histogram gradient

boosting (EXP-06) with macro-F1 = 0.5419. This is significant since it indicates that a larger search for hyperparameters did not lead to a better generalisation under the grouped validation design.

Further insight is gained from the feature ablation experiments. Demographics and context based model got the macro F1 score 0.4272 and ROC-AUC 0.6313. Inserting the early assessment information improved the macro-F1 score to 0.5214 and ROC-AUC score to 0.7576. The addition of VLE engagement, on the other hand, resulted in macro-F1 = 0.5221, and ROC-AUC = 0.7409. Consequently, both types of behavioural source make significant contributions in addition to fixed context. The results indicate that the full model, combining the assessment and engagement factors, was only slightly more predictive than either factor alone, suggesting that there is a redundant information component between the two factors related to students' progress.

The tuned gradient-boosting baseline was further improved by cohort-relative engagement to macro-F1 = 0.5455. The next decision rule experiment yielded the best conventional validation result. EXP-09 achieved macro-F1 = 0.5529, accuracy = 0.6678, and ROC-AUC = 0.7816. Its probability multipliers were (1.0, 0.7, 0.7) which, at the final operating point, favoured the majority class of the Pass class. The resulting model improved Pass recall (from 0.757 to 0.840), but decreased Fail recall (from 0.539 to 0.479) and Withdrawn recall (from 0.354 to 0.324). There was therefore no overall gain in early-warning sensitivity as a result of this improvement in macro-F1 because it was a change to a more accurate operating point.

The same performance ceiling was also observed in the deep-learning experiments. The unregularised baseline MLP had validation accuracy = 0.6897, macro-F1 = 0.5109 and a particularly low Withdrawn recall of 0.1062. Its best validation epoch is epoch 3 and then it didn't improve the chosen metric, even after training. The addition of dropout, L2 regularisation and balanced class weights yielded EXP-11, with a macro-F1 of 0.5541, a balanced accuracy of 0.5592, a ROC AUC of 0.7767 and a Withdrawn recall of 0.3811. The train-validation macro-F1 difference was also minimal (0.0278), which suggests that the model was much less prone to overfitting than the unrestricted random forest.

The performance of the hybrid LSTM-tabular model was not worse than the regularised MLP model. EXP-14 achieved validation macro-F1 = 0.5431 and ROC-AUC = 0.7785, with Withdrawn recall = 0.4132. Its highest recall was Withdrawn, but it had a fail recall of 0.4708 and an aggregate macro-F1 score below EXP-11. This means the additional information from the explicit sequence was not enough to compensate for the added complexity of the modelling. The tabular feature set already included weekly click summaries and engagement trend variables; this meant that there was reduced amount of truly new information available to the LSTM.

Table I summarises the main validation comparison.

TABLE I. SELECTED VALIDATION RESULTS

Model	Accuracy	Bal. Acc.	Macro-F1	ROC-AUC	Withdrawn Recall
Majority reference	0.586	0.333	0.246	0.500	0.000
Logistic regression, baseline	0.698	0.523	0.531	0.779	0.140
Logistic regression, weighted	0.624	0.545	0.537	0.772	0.396
Histogram gradient boosting, weighted	0.639	0.554	0.548	0.777	0.381
EXP-09, selected traditional	0.668	0.548	0.553	0.782	0.324
EXP-10, baseline MLP	0.690	0.510	0.511	0.774	0.106

Model	Accuracy	Bal. Acc.	Macro-F1	ROC-AUC	Withdrawn Recall
EXP-11, selected deep learning	0.646	0.559	0.554	0.777	0.381
EXP-14, hybrid LSTM-tabular	0.635	0.550	0.543	0.778	0.413

At the end of the test evaluation, it can be confirmed that the validation result is reasonably stable. The test macro-F1 of EXP-09 was 0.5489, which is just 0.0040 lower than the validation score. EXP-11 achieved test macro-F1 = 0.5412, a decline of 0.0129. The test ROC-AUC values were almost the same for both models (0.7788 for EXP-09 and 0.7796 for EXP-11). The test accuracy of EXP-09 was higher than EXP-11 (0.6664 vs 0.6347) while the Withdrawn recall of EXP-11 was higher than EXP-09 (0.4072 vs 0.3366). There were 4005 records from 3659 students in the test set.

TABLE II. FINAL TEST-SET PERFORMANCE

Model	Macro-F1	Accuracy	Bal. Acc.	ROC-AUC	PR-AUC	Pass Recall	Fail Recall	Withdrawn Recall
EXP-09 traditional	0.549	0.666	0.543	0.779	0.582	0.852	0.441	0.337
EXP-11 MLP	0.541	0.635	0.546	0.780	0.579	0.771	0.460	0.407

Based on the results of the bootstrap analysis, there is no significant difference in macro-F1 and ROC-AUC between EXP-09 and EXP-11 and the differences between both macro-F1 and macro-ROC-AUC are not far from zero. The Withdrawn recall is much higher in the case of EXP-11, however, showing that the two configurations encode different trade offs between general correctness and withdrawal sensitivity.

Learning curves show that the training-validation gap is narrowing but not closing with increasing data size, suggesting that more data will help, but there are still substantial ambiguities in the outcome and quality of features. There is significant difference both within and between modules (from 0.491 for module DDD to 0.000 for module GGG) and between age bands and disability statuses.

Lastly, permutation importance indicates that features that are cohort-relative have the greatest reduction in macro-F1 score when permuted. This big decrease is indicative of high model dependence and feature redundancy, however, not exclusive and independent information content.

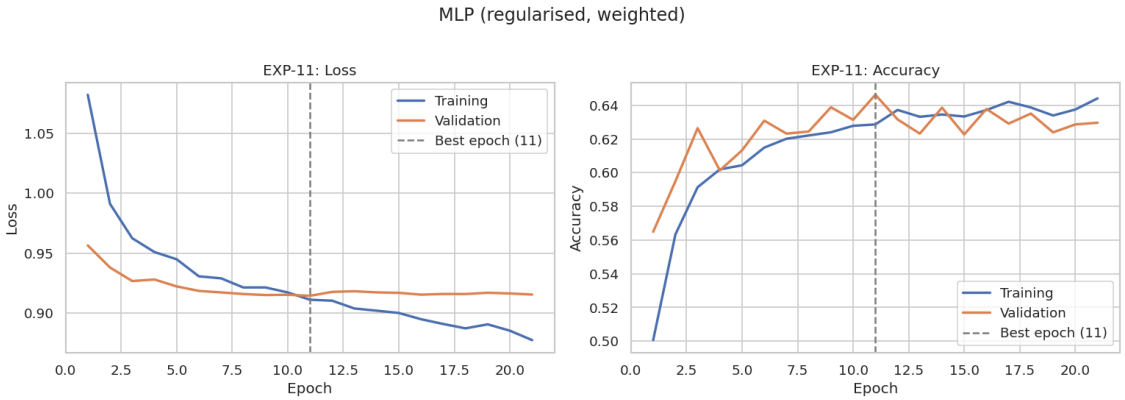


Fig. 1. Learning curves for EXP-11, the selected regularised and class-weighted MLP. Training and validation loss/accuracy are shown by epoch; the dashed marker identifies the best validation epoch.

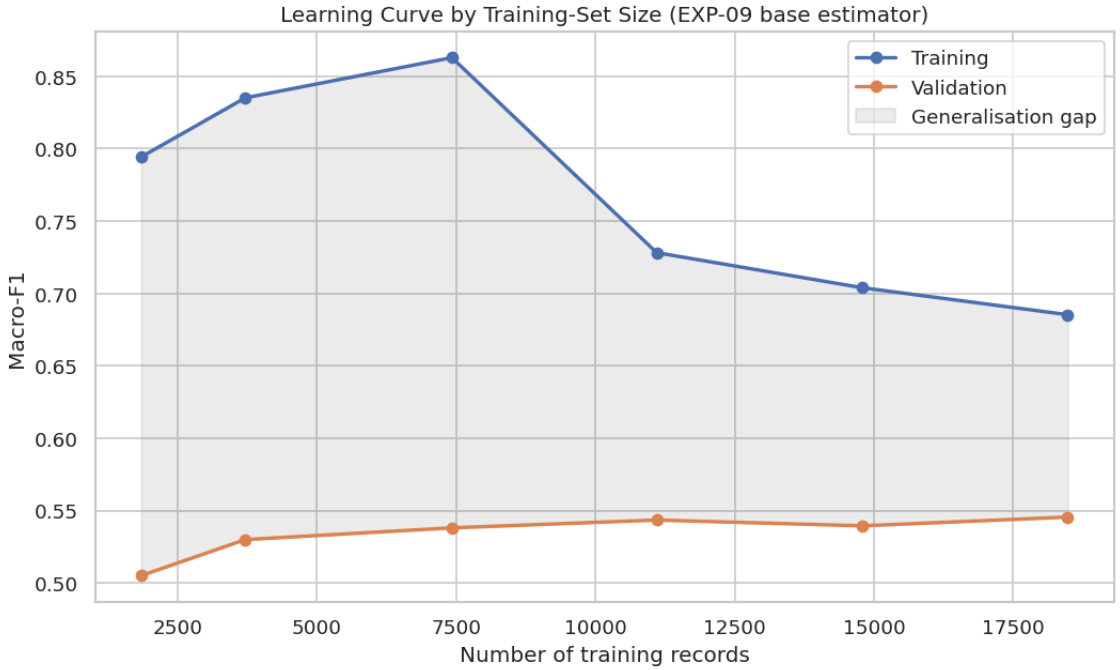


Fig. 2. Training-size learning curve for the selected traditional model base estimator. The persistent train-validation gap indicates remaining variance and/or limits in generalisation.

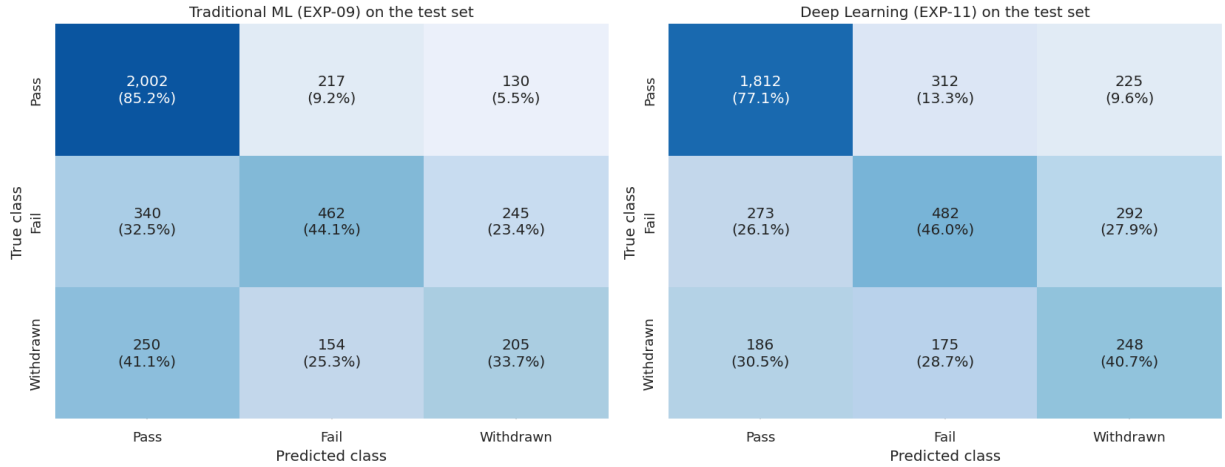


Fig. 3. Test-set confusion matrices for the selected traditional model (EXP-09) and selected deep-learning model (EXP-11). Cell annotations show counts and row-normalised percentages.

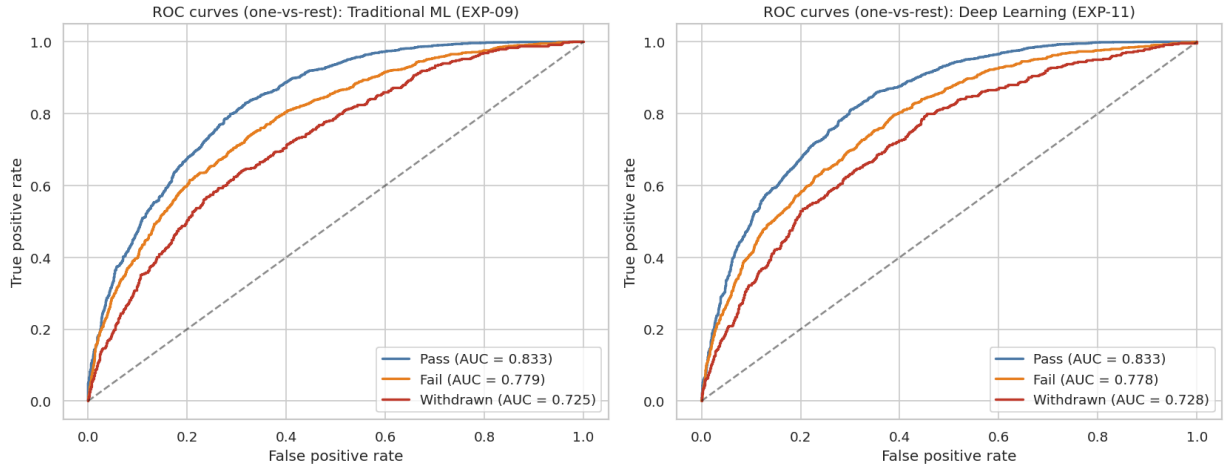


Fig. 4. One-vs-rest ROC curves on the held-out test set for the two selected models.

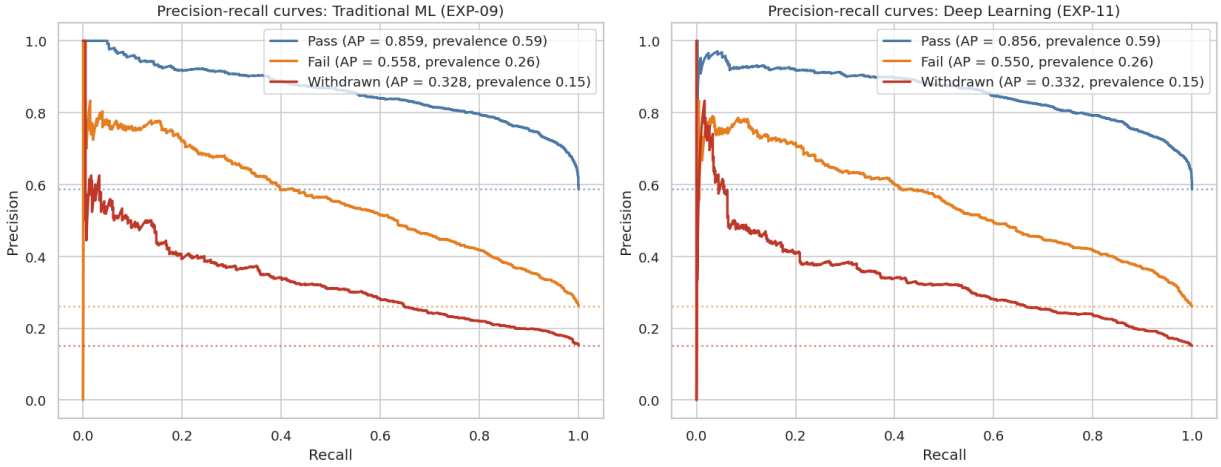


Fig. 5. One-vs-rest precision-recall curves on the held-out test set. These curves highlight the precision-recall trade-off for the minority classes.

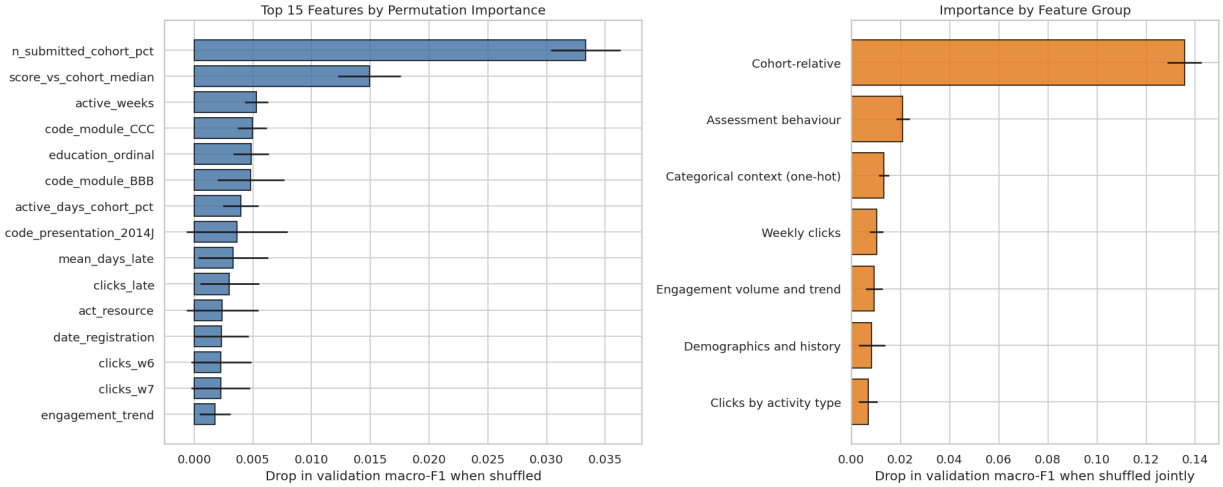


Fig. 6. Permutation importance for the selected traditional model. The left panel shows individual features; the right panel shows grouped feature-family importance.

V. Discussion

The evaluation shows that it is possible to predict the three-way student outcomes at the 8th week, albeit with some difficulty; models have an ROC-AUC of around 0.78. If performance levels are the same across all architectures (logistic regression, gradient boosting, MLPs, and LSTMs), then it suggests that the problem is not a lack of algorithmic power, but rather that the first eight weeks provide overlapping and incomplete information about the eventual outcome of a student.

Ablation analysis further shows the importance of behavioural and assessment data; the models that take only demographic data do not perform as well. But as the representation becomes more elaborate (such as LSTMs), the gains start to be less noticeable than with simple tabular features. Specific models can be compared to reveal important deployment trade-offs. Although EXP-09 can achieve more accuracy and macro-F1, bootstrap analysis reveals that the benefit is mainly in certain metrics, whereas other combinations such as EXP-11 have better results for "Withdrawn" recall.

The accuracy of the withdrawal prediction (between 0.32 and 0.35) implies that approximately two-thirds of withdrawn students would not be classified as withdrawn. These are reasonable numbers for an invitation to check-in for someone who is supportive, or for academic advising, but they are completely unacceptable for automatic administrative, financial or punitive actions. In addition, there is also remarkable variation across modules, with a 0.491 recall achieved in one course compared with 0.000 in another, indicating that a summary statistic may hide a deep performance disparity between courses and thus requiring careful calibration prior to deployment.

Error structures indicate a task might be better construed as being hierarchically organized between general risk of non-completion and specific error or withdrawal trajectories. Further, by keeping probability outputs intact instead of truncating them to hard labels, institutions can set their decision thresholds as needed in relation to their intervention capacity and resources.

The study has some definite shortcomings. The eight week time frame does not include early withdrawals, which make up about 60% of all withdrawals by the date that students unregistered, and OULAD data is based on a specific historical distance learning environment. Furthermore, the effects of unmeasured contextual factors on behavioural traces. In contrast, methodological advantages are that the student-grouped splits prevent data leakage, that both training-only imputation and scaling are adapted, that paired bootstrap analyses are used, and that a robust set of reproducibility artifacts provides zero probability discrepancy when reloading.

Finally, this model should be an integral and part of a human-in-the-loop early-warning process. The system should show probability scores rather than binary labels such as “Fail” or “Withdrawn” and provide recent engagement summaries and risk factors, which allow staff to prioritise reviews. It is important to monitor the calibration, subgroup metrics, module level performance and distributional drift continuously to ensure that retraining is undertaken responsibly.

VI. Conclusion

This study proposed a leakage-aware early-warning classification framework predicting Pass, Fail and Withdrawn grades based on the information that can be gained throughout the first eight weeks of a single module of an OULAD. The analysis involved comparison of traditional tabular machine-learning approaches with neural architectures and assessed them following a train/validation/test setup with student groups.

The best traditional model, EXP-09, achieved test macro-F1 = 0.5489, accuracy = 0.6664, and ROC-AUC = 0.7788. The best deep-learning model (EXP-11) obtained the following test scores: macro-F1 = 0.5412, accuracy = 0.6347, and ROC-AUC = 0.7796, and it had a higher recall (0.4072 versus 0.3366) for the detection of Withdrawn students. There were no statistically reliable differences between models for macro-F1 and ROC-AUC, however the paired bootstrap did provide some support for the higher value of the neural model for Withdrawn recall and for the higher value of the traditional model for accuracy.

The experiments show that the assessment and VLE engagement information are much more predictive than just demographics and context. Despite these complications, there was no clear winner of the hybrid network (LSTM-tabular) and the extensive tuning of the gradient-boosting model did not surpass the simpler version with the weighted configuration. Some care is thus indicated in feature engineering, leakage control, class-aware evaluation and operating-point selection rather than for complexity, which is why the evidence supports the former choice.

The model should be used to support, not replace, the decision-making process, and is intended as an early-warning system only. Human review is vital for moderate precision or variation in the minority class and limited temporal horizon. Further research should explore module specific calibration, hierarchical failure/withdrawal modelling, richer temporal representations, fairness and intersectional auditing, probability calibration and prospective evaluation of the effectiveness of model-supported interventions at improving student outcomes.

In general, it shows that an easily replicable eight-week prediction pipeline can give useful early indications based on educational data, but that uncertainty and limitations of the predictions can be made explicit. The predictive performance, methodological safeguards and deployment-aware evaluation offered by this combination are more useful for an early-warning system than an overall accuracy score.

References

[1] J. Kuzilek, M. Hlosta, and Z. Zdrahal, "Open University Learning Analytics dataset," *Scientific Data*, vol. 4, no. 1, pp. 170171-, Nov. 2017, doi: 10.1038/sdata.2017.171.

[2] "Penetrating the Fog: Analytics in Learning and Education," EDUCAUSE Review. Accessed: Oct. 19, 2026. [Online]. Available: https://er.educause.edu/articles/2011/9/penetrating-the-fog-analytics-in-learning-and-education

[3] "Predictive Modelling in Learning Analytics: A Machine Learning Approach in R." Accessed: Oct. 19, 2026. [Online]. Available: https://ouci.dntb.gov.ua/en/works/loeqnNW7/

[4] D. Ifenthaler and J. Y.-K. Yau, "Utilising learning analytics to support study success in higher education: a systematic review," *Educational Technology Research and Development*, vol. 68, no. 4, pp. 1961–1990, Jun. 2020, doi: 10.1007/s11423-020-09788-z.

[5] F. J. García-García, M. I. Gómez-Núñez, and C. Molla-Esparza, "Applications of learning analytics in the study of academic performance in higher education: A meta-review with an educational perspective," *Higher Education*, pp. 1–29, Jun. 2026, doi: 10.1007/s10734-026-01709-y.

[6] G. Ramaswami, T. Susnjak, A. Mathrani, and R. Umer, "Use of Predictive Analytics within Learning Analytics Dashboards: A Review of Case Studies," *Technology, Knowledge and Learning*, vol. 28, no. 3, pp. 959–980, Aug. 2022, doi: 10.1007/s10758-022-09613-x.

[7] A. Bagale, "Early Warning System for Student Dropout Risk Using Behavioral Analytics, Explainable AI, and Hybrid...," International Journal for Research in Applied Science and Engineering Technology. Accessed: Oct. 19, 2026. [Online]. Available: https://www.researchgate.net/publication/405036511_Early_Warning_System_for_Student_Dropout_Risk_Using_Behavioral_Analytics_Explainable_AI_and_Hybrid_Machine_Learning_Models

[8] C. J. Arizmendi *et al.*, "Predicting student outcomes using digital logs of learning behaviors: Review, current standards, and suggestions for future work," *Behavior Research Methods*, vol. 55, no. 6, pp. 3026–3054, Aug. 2022, doi: 10.3758/s13428-022-01939-9.

[9] S. Alturki, I. Hulpuş, and H. Stuckenschmidt, "Predicting Academic Outcomes: A Survey from 2007 Till 2018," *Technology, Knowledge and Learning*, vol. 27, no. 1, pp. 275–307, Sep. 2020, doi: 10.1007/s10758-020-09476-0.

Appendix A. Reproducibility and Saved Artifacts

The selected traditional model was saved and reloaded as a JOBLIB bundle, the selected deep-learning model was loaded in its native Keras format with an H5 copy, and a preprocessing bundle with the scaler, imputation rules, cohort references, class weights and sequence statistics were stored and loaded.

The notebook verified the saved artifacts by making comparisons to the prediction before and after loading. For both of the models, the greatest possible difference in probabilities was 0.00e+00. The generated requirements file records NumPy 2.1.3, pandas 2.2.3, SciPy 1.16.3, scikit-learn 1.6.1, Matplotlib 3.10.0, seaborn 0.13.2, joblib 1.6.0, TensorFlow 2.21.0, and TensorBoard 2.21.0.

The public project links should point to the same version of the notebook, saved model artefacts, requirements file, figures, and demonstration video used for assessment. The three project URLs are intentionally left as explicit placeholders because no public GitHub, Google Drive, or video URLs were present in the supplied notebook.